

Instructions:

- You may use any calculator except a TI-nspire.
- Put away all phones, laptops, and course materials (other than your formula sheet).
- Show all of your work, and use correct notation. Full credit will not be given without sufficient supporting work.
 - If you use a theorem, rule, definition, etc., tell me that you're using it!
- Round all numeric answers to at least 2 decimal places.
- You are awesome, and I believe in you!

1. (14 pts.) Let Y be a continuous random variable with the following pdf:

$$f(y) = \begin{cases} 3y^2, & 0 \leq y \leq 1 \\ 0, & o.w. \end{cases}.$$

- (a) (8 pts.) Find $F(y)$.

- (b) (6 pts.) Calculate $P(0.4 \leq Y \leq 0.7)$.

2. (8 pts.) A discrete random variable Y has the following cumulative distribution function:

$$F(y) = \begin{cases} 0, & y < 1 \\ 0.3, & 1 \leq y < 6 \\ 0.5, & 6 \leq y < 10 \\ 0.9, & 10 \leq y < 12 \\ 1, & y \geq 12 \end{cases}.$$

Find the probability distribution, $p(y)$, of Y .

3. (16 pts.) According to the *National Health and Nutrition Examination Survey*, published by the National Center for Health Statistics, the serum (noncellular portion of blood) total cholesterol level of U.S. females 20 years or older is normally distributed with a mean of 206 milligrams per deciliter (mg/dL) and standard deviation of 44.7 mg/dL.

(a) (8 pts.) What proportion of U.S. females 20 years or older have cholesterol levels below 200?

(b) (8 pts.) How high must the cholesterol level of a U.S. female 20 years or older be in order to fall in the highest 8%?

4. (8 pts.) Consider the random variable Y , which has a probability density function given by

$$f(y) = \begin{cases} ky^2 e^{-2y}, & y > 0 \\ 0, & o.w. \end{cases}.$$

Find the value of k that makes $f(y)$ a valid probability density function. **Hint:** integrating is not the quickest approach here.

5. (24 pts.) Suppose Y_1 and Y_2 have the following joint density function:

$$f(y_1, y_2) = \begin{cases} 3y_2, & 0 \leq y_1 \leq y_2 \leq 1 \\ 0, & o.w. \end{cases}.$$

(a) (8 pts.) Calculate $P(Y_2 \geq 2Y_1)$.

(b) (8 pts.) Find the marginal density function of Y_2 .

(c) (8 pts.) Calculate $P(Y_1 \leq 1/8 \mid Y_2 = 3/4)$.

6. (6 pts.) Identify the distribution of the random variable with the following moment-generating function. Be specific—label and give values for the parameters of the distribution.

$$m(t) = e^{20t+3t^2}$$

7. (14 pts.) Suppose X_1 and X_2 are random variables with the following properties:

$$\begin{array}{lll} E(X_1) = -2 & V(X_1) = 4 & Cov(X_1, X_2) = 2 \\ E(X_2) = 1 & V(X_2) = 3 & \end{array}.$$

- (a) (8 pts.) Let $Y = 2X_1 - 3X_2$ and $W = X_1 + X_2$. Calculate $Cov(Y, W)$.

- (b) (6 pts.) Are Y and W independent? Justify your answer.

Choose **one** of problems 8-10 to complete. If you work on more than one problem, indicate which one you want graded; otherwise, your instructor may select which one to grade.

8. (10 pts.) Prove the following result: $Cov(Y_1, Y_2) = E(Y_1 Y_2) - E(Y_1)E(Y_2)$. You may use the following notation: $\mu_1 = E(Y_1)$ and $\mu_2 = E(Y_2)$.

9. (10 pts.) Suppose $X \sim \exp(\beta)$ and $a, b > 0$. Prove that $P(X > a + b \mid X > a) = P(X > b)$.

10. (10 pts.) Let Y_1 and Y_2 be independent, $g(Y_1)$ be a function of Y_1 , and $h(Y_2)$ be a function of Y_2 . Prove the following:

$$E[g(Y_1)h(Y_2)] = E[g(Y_1)] \cdot E[h(Y_2)].$$